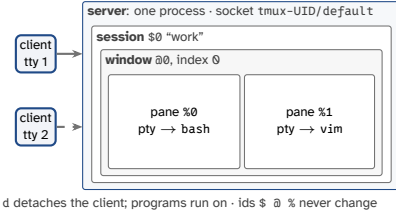
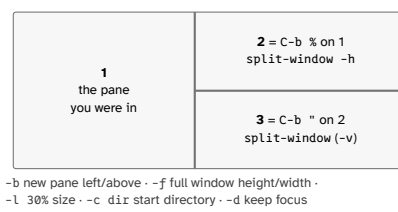


In one line: A **pane** is one pseudo-terminal running one program; panes tile a **window**, windows make a **session**, and one **server** process owns them all (socket in `$TMUX_TMPDIR` or `/tmp`), so detaching leaves every program running. Keys are the **prefix** C-b, then the key. Moving a pane moves the terminal itself — the program never restarts.

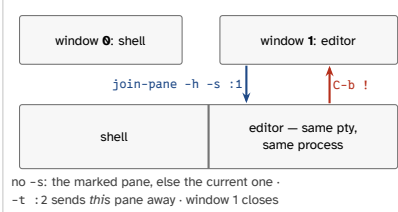
1 · server > session > window > pane



2 · a split divides the *current* pane



3 · move a pane between windows



4 · preset layouts — prefix + key; every pane stays, only the borders move (shaded = main pane)



Default keys — after the prefix C-b

split · close

% split left / right — `split-window -h`
" split top / bottom — `split-window (-v is the default)`
x kill the pane after y/n; exit in its shell (does the same)
! break the pane out into a window of its own

focus

arrows pane above / below / left / right (select-pane -U -D -L -R)
o ; next pane · last active pane (l last window, L last session)
q number every pane (display-panes); a digit jumps there

size

C-arrows resize by 1 cell · M-arrows by 5 cells (resize-pane)
z zoom: the pane fills the window, Z on its tab; again to undo

rearrange

{ } swap with the previous / next pane (swap-pane -U / -D)
C-o M-o rotate every pane forwards / backwards (rotate-window)
Space next preset layout · M-1...M-7 pick one · E even out
m M mark this pane (toggle) · clear the mark; one marked pane at a time

more

T set the pane title (3.8) · > pane menu · < window menu
* new floating pane (3.7) · g then 1-4 or an arrow: move it (3.8)
[copy mode (PgUp: enter + scroll) · paste the newest buffer
: ? command prompt · list every binding
C-b send a literal C-b — reaches a nested, inner tmux
d D detach this client · choose a client to detach
Tab fuzzy window switcher; S-Tab sessions (3.8)

r = bound with `bind -r`; repeats *without* the prefix while presses come within repeat-time (500 ms). Copy keys: vi Space begin, Enter copy; emacs C-Space, M-w.

Mouse (mouse on — the default since 3.8): click selects a pane · drag a border resizes (resize-pane -M) · drag in a pane selects in copy mode · wheel up enters copy mode unless the program is full-screen or wants the mouse · right-click: pane menu · double / triple click copies a word / line · Ctrl-drag: new floating pane (3.8).

Pane commands — the flags that matter

split-window	-h side by side · -v stacked · -b before · -f full span · -l 20 or -l 30% · -c dir · -d keep focus · -Z zoom it · -I empty pane fed from stdin
join-pane	split -t and move -s into the gap; same -h -v -b -f -l -d; the reverse of break-pane; 3.8: tiles a floating pane
move-pane	= join-pane; 3.8 adds moving floating panes
break-pane	-s pane → own window · -d stay here · -n name · -a / -b after / before this Index: 3.8: floats a tiled pane
resize-pane	-U -D -L -R n relative · -x / -y absolute, n or n% · -Z zoom · -M mouse drag · -T trim lines below cursor
swap-pane	-s -t, or -U / -D neighbour · -d keep the active pane · -Z keep zoom
rotate-window	-U / -D · -Z keep zoom
select-layout	a name, or -n / -p · -E spread this pane and its neighbours · -o undo
select-pane	-L -R -U -D · -m / -M mark / clear · -T title · -d / -e block / allow input
kill-pane	-a every pane except -t; the last pane takes its window with it
respawn-pane	rerun a dead pane's command (-k kills a live one first)
capture-pane	-p to stdout · -S / -E line range, negative = history · -J join wrapped lines · -e keep colours
pipe-pane	pipe a pane's output to a command; -o opens only if none is open, so one key toggles

What changed recently

3.5 Sep 2024	3.6 Nov 2025	3.7 Jun 2026	3.8 Sep 2026
mirrored main layouts on M-6 M-7 · escape-time 500 → 10 ms	pane scrollbars · initial-repeat-time · tiled-layout · max-columns	floating panes (new-pane, C-b *) · focus-follows-mouse	mouse on by default · Tab switcher · C-b T · JSON layout strings · break-pane floats, join-pane tiles

Targets — -t and -s

work:2.1

session: \$id · name · name prefix · glob; =name = exact only
window: token · index · @id · @id · name · prefix · glob
pane: index · %id (in \$TMUX_PANE); omitted = the active pane

window	^ lowest · \$ highest · ! previous current · + / - next / previous (+2)
pane	! last active · + / - - {top} {bottom} {left} {right} {top-left}... · - {up-of} {down-of} {left-of} {right-of} the active pane · {marked} = ~ · {mouse} =
short	:1 window 1 here · :.2 pane 2 here · :1.0 · %7 by id · work: a session

Options that shape panes

option	default	values · note
base-index	0	first window number; pane-base-index 0: first pane
synchronize-panes	off	input goes to every pane of the window not in a mode on · failed-key (3.7) · failed-key (3.8); then respawn-pane
remain-on-exit	off	
display-panes-time	1000 ms	how long the C-b q numbers stay
repeat-time	500 ms	keys; initial-repeat-time (3.6) 0 = the same
escape-time	10 ms	500 before 3.5; a 0.5 s wait after every Esc
history-limit	2000	scrollback lines per pane
main-pane-width	80	main-pane-height 24; the main pane of main-*
pane-border-status	off	top · bottom: a little line on each border
pane-border-lines	single	double · heavy · simple · number · spaces (3.6)
pane-scrollbars	off	modal · on (3.6) · auto-hide (3.8)
mouse	on	off before 3.8
focus-follows-mouse	off	hover selects the pane (3.7; needs mouse)
mode-keys	emacs	vi if \$VISUAL or \$EDITOR contains "vi"
renumber-windows	off	windows keep the gap a closed one leaves

Saving layouts: resurrect + continuum

Plugins via TPM: set -g @plugin 'tmux-plugins/tmux-resurrect', prefix I installs. **resurrect:** prefix C-s save · prefix C-r restore sessions, windows, panes and their order, layouts, working directories, the active pane; reruns only vi vim nvim emacs man less more tail top htop irssi weechat mutt; pane text with @resurrect-capture-pane-contents 'on'; files in ~/.tmux/resurrect; panes that exist are left alone. **continuum:** autosave every 15 min (@continuum-save-interval, 0 = off), @continuum-restore 'on' restores at server start. It hooks status-right: status line on, plugin listed last — a theme that rewrites status-right stops the autosave silently.

Recipes

```
# terminal on the right, 30 %, same directory
tmux split-window -h -l 30% -c '#{pane_current_path}'
# type into every pane of this window (off to stop)
tmux set -w synchronize-panes on
# last 200 lines of pane %3 as text
tmux capture-pane -p -S -200 -t %3
# log a pane to a file; the same key stops it (man page)
bind-key C-p pipe-pane -o 'cat >>~/output.##I-#P'
```

Traps

- ! -h is side by side: tmux names the split by how the panes line up, Vim names the same split :vsplit.
- ! A forgotten mark: join-pane, move-pane, swap-pane without -s take the **marked** pane — C-b M clears it.
- ! M-keys type symbols on a Mac until Option sends Meta: Terminal Use Option as Meta key; iTerm2 Option key Esc+.
- ! With mouse on, the terminal's own selection needs **Shift** (many Linux terminals) or **Option** (iTerm2).